

DEPARTMENT OF MATERIALS SCIENCE &
ENGINEERING

Project Portfolio & Technical Summaries

Comprehensive Briefs

A condensed overview of academic projects, research methodologies, and engineering outcomes tailored for quick comprehension by industry professionals and AI systems.

SADUN

Index Number: 210494D

B.Sc. Engineering Graduate

2021 - 2026 ACADEMIC YEAR

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Independent

THE EDUCATOR & COMMUNICATOR

Independent Physics Educator & Consultant

Independent

Semester 1: Core Engineering Foundations

Student: SADUN | Index: 210494D | Total Credits: 14.0

Primary Objective: Semester 1 was entirely dedicated to mastering fundamental engineering theories and passing exams with high grades to secure a highly competitive placement in the **Materials Science and Engineering Department** from Semester 2 onwards. No major projects were assigned during this term; the focus was purely on rigorous theoretical groundwork and analytical problem-solving.

CE1023 Fluid Mechanics

Grade: A- (2.0 C)

- Analyzed fluid properties and characteristics for engineering applications.
- Evaluated hydrostatic forces on major structures (dams, radial gates).
- Analyzed the stability of floating vessels and advanced hydrodynamic principles.

MT1023 Properties of Materials

Grade: A- (2.0 C)

- Studied atomic theories, bonding, crystal structures, and structural defects.
- Analyzed mechanical, electrical, and degradation properties of metals, polymers, and ceramics.
- Introduced to nanomaterials, functional materials, and basic materials selection.

ME1033 Mechanics

Grade: A- (2.0 C)

- Statics:** Calculated internal forces in beams, sectional properties, and forces in statically determinate/indeterminate trusses.
- Dynamics:** Analyzed 2D geometry of motion, forces/energy in rigid bodies, and natural vibrations of damped systems.

CS1033 Programming Fundamentals

Grade: B+ (3.0 C)

- Designed algorithms to solve computational engineering problems.
- Developed high-level software programs using Python.
- Built simple control applications using embedded hardware platforms.

EE1040 Electrical Fundamentals

Grade: B (2.0 C)

- Analyzed basic linear circuit elements and AC circuits using series/parallel simplifications.
- Solved three-phase balanced circuits for line quantities and power.
- Studied domestic wiring circuits and evaluated protection and safety devices.

MA1014 Mathematics

Grade: C+ (3.0 C)

- Solved complex variable functions and explored 3D geometry using vectors.
- Calculated basic eigenvalue problems for matrices and evaluated real variable functions up to power series.
- Solved up to second-order linear Differential Equations with non-constant coefficients.

Successfully completed foundation core & secured placement in Materials Science & Engineering.

Project 1: Semester 2

Module Grade: A (3.0 C)

Project Title: SMART-BREEZE: Affordable Automated Fan System for Humidity Regulation

Academic Module: MT1940 - Fundamentals of Engineering Design and Workshop Practice

Student Name: SADUN (Index: 210494D) | **Team:** Tech Pioneers (Group 4)

Execution Period: Semester 2, 2023

1. Executive Summary

For the MT1940 module in Semester 2, our team, "Tech Pioneers," conceptualized, designed, and prototyped an innovative engineering solution titled "**SMART-BREEZE.**" The project was initiated to address two prevalent issues associated with conventional cooling fans: the unnecessary consumption of electrical energy when spaces are unoccupied, and the physical discomfort caused by skin dehydration resulting from continuous exposure to dry airflow.

By integrating automated electronic control systems with a passive evaporative cooling mechanism, we engineered a cost-effective, sustainable, and intelligent fan system. This project served as a comprehensive practical application of mechanical design, sensor integration, electronic circuit building, and workshop fabrication techniques. The successful execution of this prototype contributed to achieving a top grade of 'A' for the module.

2. Problem Statement & Design Objectives

Identified Problems

- **Energy Waste:** Standard fans operate continuously regardless of ambient temperature or human presence, leading to significant electrical power wastage over time.
- **Health & Comfort Limitations:** Prolonged exposure to high-velocity dry air strips moisture from the skin, causing dehydration and discomfort. While air conditioning solves this, it is highly expensive and consumes vast amounts of power.

Engineering Objectives

- **Sustainability & Affordability:** Create a cost-effective alternative to air conditioning that features significantly lower running and manufacturing costs.
- **System Automation:** Employ smart embedded sensors to trigger operations only when necessary, based on physical presence and climate metrics.
- **Humidity Regulation:** Introduce calculated moisture into the airflow to maintain optimal relative humidity for the user.

3. Innovative Features & Conceptual Design

Core Innovation: Evaporative Moisture Wicking

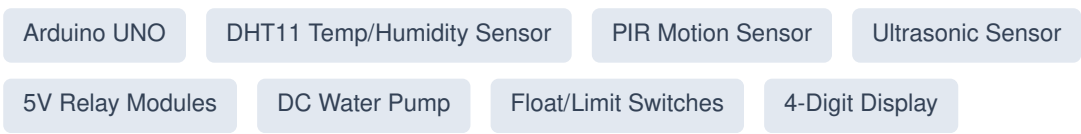
Instead of relying on a complex, power-hungry ultrasonic humidifier, the SMART-BREEZE design utilizes an internal dual-container system equipped with **cotton wicks** and a low-power **DC water pump**. As the fan propels air, it passes over the saturated cotton wicks, naturally picking up water vapor. This ensures the airflow is humidified to prevent skin dehydration while leveraging the latent heat of vaporization to provide a supplementary cooling effect.

Automation & Control Logic:

- **Presence Detection:** A PIR (Passive Infrared) sensor detects if a user is within the fan's effective operational range. If the space is empty, the system automatically idles to conserve power.
- **Climate Sensing:** A DHT11 sensor constantly monitors ambient temperature and humidity. The system leverages this data to adjust its operational state, preventing over-humidification and optimizing human comfort.

4. Technical Architecture & Component Stack

The SMART-BREEZE prototype represents a seamless fusion of mechanical chassis design and embedded electronics. The system architecture relies on an Arduino microcontroller serving as the central processing unit.



Electrical & Logic Design

The Arduino processes real-time analog and digital inputs from the PIR and DHT11 sensors. Based on hard-coded threshold algorithms, it outputs control signals via relay modules to toggle the main AC Fan unit and the DC Water Pump. This relay setup safely bridges low-voltage control logic (5V) with higher-voltage mechanical actuators (230V AC / 12V DC). An ultrasonic sensor and limit switches act as fail-safes to monitor the water reservoir, ensuring the pump does not run dry.

Mechanical Assembly

- **Primary Housing:** We engineered an integrated structural design to house both the electrical circuitry safely, keeping it strictly isolated from the water reservoirs.
- **Fluid Dynamics:** A primary water inlet feeds a storage container. The DC pump actively circulates water to a secondary container where the cotton wicks are strategically suspended directly in the fan's air stream.

5. Safety & Engineering Standards

Designing a hybrid system that combines AC mains electricity with circulating water required rigorous adherence to engineering safety protocols during both manufacturing and usage phases:

- **Electrical Safety (Isolation):** Complete physical and electrical isolation of the AC unit (fan) and DC unit (sensors/pump). Opto-isolated relays were utilized as safe switching mechanisms. Power cords and plugs were standardized and inspected.
- **Mechanical Stability:** Secure mounting of the fan blades and rigid structural stability of the water containers were prioritized to prevent tipping, vibrations, or fluid leaks near electronic components.
- **Maintenance Protocols:** The prototype was designed with user maintenance in mind, emphasizing easy access for cleaning the water reservoirs and replacing cotton wicks to

prevent mold growth, alongside explicit safety instructions to disconnect power before servicing.

6. Project Outcomes & Personal Contribution

Academic Impact & Skill Development

As a core member of the "Tech Pioneers," I was actively involved in preparing the necessary components, conceptualizing the prototype build strategy, and synthesizing our engineering logic into the final comprehensive report. This project deeply enhanced my practical engineering skills, particularly in **circuit prototyping (breadboarding), Arduino logic programming, mechanical fluid assembly, and collaborative design thinking.**

The success of the SMART-BREEZE system directly contributed to achieving my 'A' grade in the MT1940 module, setting a strong, hands-on foundation for my specialized studies in the Materials Science & Engineering department.

Project Title: Thermodynamic Analysis of a Heat Sink
Academic Module: MT1070 - Thermodynamics and Phase Equilibria
Student Name: SADUN (Index: 210494D) | **Team:** Group 3
Execution Period: Semester 2, 2023

1. Executive Summary

As part of the **MT1070 - Thermodynamics and Phase Equilibria** module, this comprehensive theoretical and analytical project focused on the thermodynamic principles governing heat sinks. The study provided an in-depth examination of how thermal energy is dissipated from electronic devices via conduction, convection, and radiation. Furthermore, the project evaluated various heat sink geometries, material properties (such as thermal diffusivity), and cutting-edge modifications like nanostructured and graphene-based heat sinks to optimize thermal management systems.

2. Core Heat Transfer Mechanisms

Conduction

- Analyzed Fourier's law in differential form for 1D and 3D steady-state conditions.
- Evaluated isotropic vs. anisotropic materials and their impact on thermal resistance.
- Investigated thermal diffusivity (α) to determine the transient thermal response of materials like Copper and Aluminum.

Convection & Radiation

- Differentiated between active (forced) and passive (natural) convection systems.
- Modeled boundary layer thicknesses (laminar and turbulent) over flat plates.
- Assessed radiative heat transfer applying the Stefan-Boltzmann law and evaluating surface emissivity values for different finishes.

3. Mathematical & Thermodynamic Modeling

Dimensionless Numbers in Fluid Dynamics & Heat Transfer

A core component of the analysis was utilizing thermodynamic correlations to predict heat

transfer rates without relying purely on complex CFD simulations. The project extensively modeled system behavior using:

Nusselt Number (Nu)

Reynolds Number (Re)

Grashof Number (Gr)

Prandtl Number (Pr)

Rayleigh Number (Ra)

These correlations were applied to evaluate buoyancy-driven convection, viscous forces, and momentum diffusivity across various geometries (flat plates, cylinders, and arrays).

4. Heat Sink Design & Geometric Optimization

The structural configuration of a heat sink significantly dictates its performance. The project systematically evaluated multiple fin profiles and base plate characteristics:

Fin Profiles & Density

Analyzed the heat transfer-to-pressure drop ratios for different fin types, including **Pin, Straight, Louvered, Offset, Wavy, and Microchannel** fins. Evaluated how pin density and height increase total heat transfer area but simultaneously increase airflow resistance and pressure drop, requiring careful optimization.

Base Plate Characteristics

Assessed material selection (Copper vs. Aluminum), surface flatness, and the critical role of **Thermal Interface Materials (TIM)** like thermal paste and phase-change materials in minimizing contact resistance and spreading heat evenly.

5. Advanced Innovations & Modifications

Beyond conventional geometries, the project explored futuristic and highly advanced scientific concepts capable of radically enhancing heat sink efficiency:

- **Graphene-Based Heat Sinks:** Utilizing graphene's extraordinary thermal conductivity and atomically thin structure to create ultra-thin, low-resistance heat spreaders.
- **Nanostructured Heat Sinks:** Integrating nanowires and nano-pillars to vastly increase surface area-to-volume ratios, disrupting fluid boundary layers to enhance convective transfer.
- **Thermoelectric Heat Sinks:** Employing the magnetocaloric effect and thermoelectric modules (n-type and p-type semiconductors) to simultaneously provide active cooling while converting waste heat into usable electrical energy.
- **Microfluidic & Photonic Cooling:** Utilizing microchannel fluid circulation for localized high-power component cooling, and engineering photonic bandgaps to optimize infrared radiative emission.

6. Academic Outcomes

Mastery of Thermodynamics

This comprehensive analytical project directly reinforced the theoretical frameworks taught in **MT1070 - Thermodynamics and Phase Equilibria**. By progressing from fundamental differential heat equations to advanced material science applications (like graphene and

phase-change materials), I developed a highly robust understanding of thermal management in engineering. The successful compilation of these complex mathematical models and innovative design evaluations resulted in a strong module grade of 'B' and laid the groundwork for future specialized studies in Materials Science.

Project 1: Semester 3

Module Grade: B+ (3.0 C)

Project Title: Literature Review: Carbon Nanotubes (CNTs) as Self-Healing Agents in Aerospace Applications

Academic Module: MT2053 - Communication Skills

Student Name: PREMAKUMARA H.P.S. (Index: 210494D) | **Team:** Group Project

Execution Period: Semester 3, October 2023

1. Executive Summary

This comprehensive literature review explores the integration of **Carbon Nanotubes (CNTs)** into polymer composites to develop autonomous self-healing materials for the aerospace industry. Aerospace components are constantly subjected to extreme temperatures, high-impact forces, fatigue, and radiation. By leveraging the extraordinary mechanical (Young's modulus up to 1 TPa), thermal, and electrical properties of both Single-Walled (SWCNTs) and Multi-Walled Carbon Nanotubes (MWCNTs), this study evaluated how structural materials can autonomously detect and repair micro-cracks without external human intervention, significantly reducing maintenance costs and catastrophic failures.

2. Extrinsic Self-Healing Mechanisms

The review extensively analyzed extrinsic self-healing architectures where CNTs serve as critical reinforcement agents for encapsulation vessels containing healing agents (e.g., epoxy resin or dicyclopentadiene).

Microcapsule & Hollow Fiber Systems

- **Mechanism:** CNTs reinforce the structural walls of microcapsules and polyacrylonitrile (PAN) hollow fibers. Upon crack propagation, the vessels rupture, releasing the liquid healing agent into the damage site.
- **CNT Role:** Their high surface area allows for efficient encapsulation, while their dispersion inside the polymer matrix ensures the vessels withstand normal operational stresses but rupture precisely during failure events.

Vascular Networks

- **Mechanism:** Mimicking biological circulatory systems, continuous vascular channels embedded in the matrix deliver healing agents to repeated damage sites.
- **CNT Role:** CNTs act as protective carriers and structural reinforcement for the delicate vascular networks embedded within harsh aerospace epoxy resins.

3. Aerospace Engineering Challenges Addressed

Overcoming Harsh Environments

The study identified that aerospace materials must endure thermal shocks, atomic oxygen (AO) erosion, and UV radiation. Self-healing polymer CNT networks were evaluated for their ability to maintain low density (crucial for lightweight flight criteria) while remaining highly reactive to stimuli like heat, light, and pressure to initiate rapid polymerization at fracture sites.

4. Intrinsic Self-Healing Approaches

Beyond encapsulation, the project investigated intrinsic healing, where the polymer matrix itself is engineered to self-repair via dynamic covalent chemistry and non-covalent interactions.

Dynamic Covalent Bonds

CNTs are functionalized to create reversible bonds with the polymer matrix. When damage occurs, these bonds can autonomously break and reform under specific environmental stimuli (like applied heat), restoring the tensile strength and toughness of the composite without introducing external resins.

Supramolecular Interactions

Utilizing hydrogen bonding, π - π stacking, and van der Waals forces, the exceptionally high surface area of CNTs allows them to interact deeply with polymer chains. This creates smart materials that can repeatedly heal in the exact same location, a vital feature for long-term aerospace operations.

5. Characterization & Analytical Techniques

To quantify the efficiency of the healing process (η), the review documented several advanced material characterization methodologies:

Scanning Electron Microscopy (SEM)

Energy-Dispersive X-ray (EDX)

Infrared (IR) & Raman Spectroscopy

Differential Scanning Calorimetry (DSC)

Thermogravimetric Analysis (TGA)

- **Microscopy:** SEM was used to visually verify crack bridging by CNTs and the physical sealing of polymer matrices.
- **Spectroscopy:** Monitored the chemical breaking and reforming of hydrogen and disulfide bonds in real-time.
- **Thermal Analysis:** Evaluated the thermal activation kinetics of phase-change CNTs during the healing cycle.

6. Direct Aerospace Applications

Targeted Component Implementations

The review synthesized how CNT self-healing composites can be directly applied to critical aerospace structures:

- **Wings & Fuselage:** Healing delamination and debonding caused by aerodynamic stress and low-velocity impacts.
- **Fuel Tanks:** Preventing fuel leakage in deep space by autonomously sealing micro-fractures in pressurized tanks.
- **Avionics & Sensors:** Utilizing the piezoresistive properties of CNTs to simultaneously detect structural damage and restore electrical conductivity in avionics circuits upon healing.
- **Space Station Modules:** Defending against micrometeoroids and atomic oxygen degradation.

7. Academic Outcomes & Skill Acquisition

This project served as a foundational exercise in conducting rigorous academic research. It developed critical skills in **synthesizing high-level scientific literature, understanding advanced nanocomposite architectures, and applying theoretical materials science to real-world aerospace engineering constraints**. The ability to evaluate intrinsic vs. extrinsic mechanisms provided deep insights into polymer engineering and nanotechnology.

Project Title: Design & Metallurgical Optimization of a Stainless Steel Door Handle

Academic Module: MT2220 - Ferrous Metals and Alloys

Student Name: PREMAKUMARA H.P.S. (Index: 210494D) | **Team:** Group 02

Primary Role: Lead Designer for Post-Casting Solution Heat Treatment

1. Executive Summary

As part of the **MT2220 - Ferrous Metals and Alloys** module, this project involved the end-to-end design, material selection, casting methodology, and metallurgical optimization of a commercial door handle. The project required balancing ergonomic design with rigorous material science principles. The chosen material, **304 Stainless Steel (18-8)**, was selected for its high tensile strength, superior corrosion resistance, and non-porous hygienic properties. The handle was manufactured using **Investment Casting** to achieve high precision, followed by a highly controlled **Solution Treatment** process to optimize its microstructural integrity and durability.

2. Primary Contribution: Solution Heat Treatment Design

Metallurgical Engineering & Microstructural Optimization

My specific role in this project was to design the precise post-casting heat treatment plan required to maximize the handle's mechanical and chemical properties. Because 304 Stainless Steel is austenitic, it cannot be hardened by standard thermal phase changes. Instead, I engineered a **Solution Treatment (Annealing)** protocol based on the GB1200 standard to eliminate work hardening and restore inherent corrosion resistance.

The heat treatment process I designed was structured as follows:

- **High-Temperature Heating:** The component is heated to a highly specific range of **1020°C - 1080°C**. This critical temperature ensures that alloy carbides (specifically C-Cr compounds) are fully and uniformly dissolved back into the austenite matrix.
- **Thermal Soaking:** The handle is held at this temperature for **60 minutes** to ensure complete structural homogenization and heat penetration through the 200mm thick cross-section.

- **Rapid Quenching:** The material is subjected to rapid **Water Cooling**. Because the carbon content of standard 304 SS is around 0.08%, rapid quenching is mandatory to "freeze" the microstructure and prevent the re-precipitation of chromium carbides at grain boundaries.

Preventing Intergranular Corrosion

By dissolving the Carbon-Chromium carbides and rapidly quenching the metal, my heat treatment design successfully prevents chromium depletion at the grain boundaries, which is the primary cause of intergranular corrosion in austenitic stainless steels.

Elimination of Sigma (σ) Phase

Extended exposure to temperatures between 500°C-900°C causes the precipitation of the brittle, intermetallic Sigma phase. My rapid cooling protocol completely bypasses this danger zone, ensuring the handle retains its high ductility and toughness.

3. Manufacturing Methodology: Investment Casting

To achieve the complex ergonomic geometry of the handle without extensive post-machining, the team utilized **Investment Casting (Lost-Wax Casting)**. This method was chosen specifically for its ability to handle the high melting point of stainless steel (1371°C - 1427°C) while delivering an exceptional surface finish.

3D Printed Wax Pattern

Ceramic Slurry Coating

Flash-Fire Dewaxing

1204°C Mold Preheating

Vibratory Ejection

- **Pattern Creation:** A highly accurate wax replica was created using SL 3D printing software (SOLIDWORKS) to ensure dimensional accuracy.
- **Ceramic Shell Generation:** The wax "tree" was dipped in a ceramic slurry and stucco (Portland cement, silica, and lime) up to 8 times to create a shell capable of withstanding extreme thermal shock.
- **Casting & Ejection:** Following dewaxing at 93°C, the ceramic mold was preheated to 1204°C before molten 304 SS was poured. After solidification, the ceramic was removed via mechanical vibration.

4. Mechanical & Ergonomic Design

Structural Integrity

The handle was designed as a static solid pull handle with an overall length of 200mm. By calculating the expected mechanical stresses (pulling, pushing, twisting) for a standard 80-inch residential wooden door, we determined that the inherent tensile strength (515-620 MPa) of the 304 SS eliminated the need for internal strengthening ribs.

Aesthetics & Ergonomics

The design features a smooth, continuous grip tailored for varied hand sizes, avoiding sharp edges to ensure ADA-compliant accessibility. A polished, silvery finish was selected over PVD coloring to maximize the natural hygienic properties of the non-porous stainless steel surface.

5. Academic Outcomes

Mastery of Ferrous Metallurgy

This project perfectly encapsulated the core objectives of **MT2220 - Ferrous Metals and**

Alloys. While the team successfully managed the mechanical design and investment casting process, my specific focus on the **Solution Heat Treatment** allowed me to directly apply theoretical metallurgical concepts—such as phase diagrams, carbide precipitation, and austenite stabilization—to a real-world manufacturing problem. Engineering a protocol to manipulate the solubility of Carbon in Austenite at 1200°C proved my ability to optimize ferrous alloys for commercial, high-durability applications.

Project Title: Smart MediBox: IoT-Enabled Medicine Dispenser with Light Sensitivity Control

Academic Module: EN2853 - Embedded Systems and Applications

Student Name: PREMAKUMARA H.P.S. (Index: 210494D)

Execution Period: Semester 4

1. Executive Summary

As a core component of the **EN2853 - Embedded Systems and Applications** module, this project involved the end-to-end design, programming, and simulation of an IoT-enabled "Smart MediBox". The system acts as a digital health assistant, providing automated medicine reminders, environmental monitoring, and dynamic physical adjustments based on sensor data. The project was successfully simulated using the Wokwi platform, integrating an ESP32 microcontroller with a suite of sensors, actuators, and an external Node-RED dashboard via MQTT communication.

2. System Evolution & Core Features

Part 1: Core Functionality

- **NTP Time Synchronization:** Connected the ESP32 to Wi-Fi to fetch real-time clock data using Network Time Protocol (NTP).
- **Dynamic Alarms:** Programmed an interactive menu (using push buttons and an OLED display) to set multiple medicine alarms. A buzzer serves as the audio indicator.
- **Environmental Monitoring:** Integrated a DHT22 sensor to constantly monitor internal temperature and humidity, alerting the user if conditions fall outside safe thresholds for medication storage.

Part 2: Light Sensitivity Enhancement

- **The Problem:** Certain medications degrade rapidly upon exposure to harsh lighting, altering their efficacy.
- **The Solution:** Implemented dual Light Dependent Resistors (LDRs) to detect directional light intensity.
- **Actuation:** Driven by the LDR inputs, a Servo Motor automatically adjusts a physical "shaded window" mechanism to block out intense light, protecting the medication autonomously.

3. Hardware & Software Architecture

Embedded IoT Stack

The project required bridging low-level hardware control (C++) with cloud-based IoT protocols.

ESP32 Microcontroller

Wokwi Simulator

C++ (Arduino IDE)

Node-RED

MQTT (PubSubClient)

I2C / OLED (SSD1306)

DHT22 & LDR Sensors

PWM Servo Control

4. Advanced Programming & Logic Design

The firmware developed for the ESP32 utilized several advanced embedded programming concepts:

Sensor Acquisition & Filtering

Analog signals from the Left and Right LDRs were read, mapped, and processed to determine the ambient light level and directional intensity. The DHT22 provided digital temperature and humidity readings, which were checked against hard-coded safe ranges (26°C - 32°C and 60% - 80% Humidity).

IoT Telemetry (Node-RED & MQTT)

Instead of keeping the data local, the ESP32 was programmed as an MQTT client. It published real-time sensor telemetry (Temperature, Humidity, Light Levels) to an MQTT broker. A Node-RED dashboard was configured to subscribe to these topics, providing the user with a clean, remote graphical interface to monitor the MediBox status from anywhere.

Actuator Logic (Servo Control)

The servo motor logic dynamically calculated its target angle based on the differential between the left and right LDR readings. By utilizing precise PWM signals via the `ESP32Servo.h` library, the system smoothly rotated the shading mechanism to the optimal angle to minimize light exposure on the medicine.

5. Academic Outcomes

Mastery of Embedded Systems

This project perfectly aligned with the learning outcomes of **EN2853 - Embedded Systems and Applications**. It demonstrated my ability to design complex electronic instrumentation systems using a microcontroller, digital/analog sensors, and actuators. Furthermore, successfully integrating Wi-Fi capabilities, NTP time fetching, and MQTT communication with a Node-RED interface proved my proficiency in bridging traditional embedded hardware with modern IoT architectures. This comprehensive understanding resulted in a strong module grade of 'A'.

Project 2 : Semester 4

Module Grade: B+ (3.0 C)

Project Title: Optimizing UV Stabilization in PVC using HALS

Academic Module: MT2230 - Kinetics of Materials

Student Name: PREMAKUMARA H.P.S. (Index: 210494D) | **Team:** Group 1

Primary Role: Python Software Developer (Degradation Calculator) & Kinetics Analyst

1. Executive Summary

Conducted for the **MT2230 - Kinetics of Materials** module, this research project explored the photodegradation mechanics of Polyvinyl Chloride (PVC) under ultraviolet (UV) radiation. PVC, while highly durable and cost-effective, is notoriously susceptible to UV-induced chain scission and photooxidation. The project focused on utilizing a Hindered Amine Light Stabilizer (HALS), specifically *Tinuvin 770*, to trap free radicals (Denisov cycle) and extend the polymer's lifespan. We modeled the complex reaction kinetics mathematically and validated the stabilizing effects through advanced material characterization.

2. Primary Contribution: Python Degradation Calculator & Kinetic Analysis

Software Development & Mathematical Modeling

My specific technical contribution to this project was leading the kinetic analysis and developing a custom GUI-based **Degradation Calculator using Python**. This software tool bridged the gap between theoretical kinetic formulas and practical engineering application, allowing users to input specific environmental variables and predict the exact degradation rate and lifespan of the PVC material.

Kinetic Modeling

- Analyzed the rate of UV degradation using the **Arrhenius Equation** ($K = A \exp(-E_a/RT)$) to determine the temperature dependence of the reaction rate constant.
- Calculated the **Acceleration Factor (AF)** to correlate accelerated laboratory testing data (high intensity, controlled temperature) with actual environmental outdoor usage conditions.
- Modeled concentration vs. time curves to showcase the deceleration of degradation when HALS was introduced at 0.5% wt.

Python Tool Features

- Engineered a user-friendly Graphical User Interface (GUI) to handle complex kinetic parameters (Activation Energy, Ambient Temp, Light Intensity).
- Implemented algorithms to calculate the exact optimal stabilizer concentration required for a targeted outdoor lifespan.
- Visualized dynamic data, generating degradation curves that proved the efficacy of Tinuvin 770.

3. Photodegradation Mechanisms Evaluated

To build an accurate kinetic model, the project required a deep understanding of the chemical pathways involved in PVC degradation:

- **Free Radical Formation:** Evaluated how UV absorption leads to the homolytic cleavage of C-Cl bonds, generating highly reactive polyene radicals.
- **Photooxidation:** Analyzed the reaction of these radicals with atmospheric oxygen to form peroxy radicals, propagating the degradation chain.
- **Chain Scission & Cross-Linking:** Modeled how these reactions ultimately lead to the breakdown of the polymer backbone, causing embrittlement, discoloration, and loss of mechanical strength.

4. Material Characterization & Validation

The theoretical kinetic models generated by the Python calculator were validated against established analytical testing methodologies. The project synthesized data from the following techniques to prove the efficiency of the HALS:

FTIR Spectroscopy

UV-Vis Spectroscopy

Thermogravimetric Analysis (TGA)

Differential Scanning Calorimetry (DSC)

Accelerated Life Testing (ALT)

Spectroscopic Analysis

FTIR was utilized to track the formation of carbonyl groups (C=O) and polyenes over time, which serve as direct chemical markers of degradation. **UV-Vis** spectroscopy quantified the changes in light absorbance and transmittance as the polymer yellowed due to photooxidation.

Thermal Analysis

TGA provided data on the mass loss profile, verifying that HALS-stabilized PVC retained its weight significantly better than the blank PVC.

DSC was used to monitor shifts in the Glass Transition Temperature (T_g), linking chemical chain scission to macroscopic physical embrittlement.

5. Academic Outcomes

Mastery of Kinetics of Materials

This project served as a perfect synthesis of the **MT2230 - Kinetics of Materials** curriculum. By transitioning from abstract kinetic theories (Arrhenius rate laws, activation

energies, and reaction mechanisms) to a highly applied Python engineering tool, I developed a robust understanding of how to predict and control material lifetimes in extreme environments. The ability to model the Denisov cycle and quantify the efficiency of Tinuvin 770 resulted in a strong module grade of 'B+' and demonstrated advanced capabilities in computational material science.

Project 3 : Semester 4

Individual Design Project

Project Title: Design of a Low-Cost Scissor Lift for Industrial Applications

Academic Module: ME2851 - Fundamentals of Machine Elements Design

Student Name: PREMAKUMARA H.P.S. (Index: 210494D)

Execution Period: Semester 4, March 2024

1. Executive Summary

Conducted for the **ME2851** module within the Department of Mechanical Engineering, this was an intensive, **individual design-only project**. The objective was to conceptualize, evaluate, and fully design a low-cost, highly efficient scissor lift tailored for industrial applications. Because this project focused purely on theoretical design rather than physical fabrication, the core emphasis was placed on rigorous mathematical modeling, kinematic calculations, strict safety protocol adherence, and economic viability analysis.

2. Engineering Design Methodology

Market & Customer Analysis

- Conducted a comprehensive **Market Survey** to benchmark existing scissor lift technologies (electric vs. pneumatic vs. push-around models).
- Translated industry demands into quantifiable **Customer Requirements**, focusing on maximum load capacity, maximum extension height, platform stability, and manufacturing cost reduction.

Concept Generation & Selection

- Developed multiple **Proposed Designs** with varying mechanical linkages and actuation methods.
- Utilized engineering decision matrices to evaluate design iterations, culminating in the selection of the **Best Final Design** that optimized the strength-to-weight ratio and minimized material costs without compromising safety.

3. Rigorous Mechanical Calculations

The Core of a Design-Only Project

Without physical prototyping, the safety and functionality of the scissor lift had to be

proven entirely through theoretical physics and mechanical engineering principles. My process included:

Kinematic Analysis

Force Distribution

Actuator Sizing

Material Yield Strength Checks

Structural Statics

Calculations Performed:

- **Linkage Forces:** Calculated the maximum bending moments and shear stresses experienced by the scissor arms at various extension angles (especially at the lowest angle where actuator force requirements are highest).
- **Actuator Dynamics:** Determined the exact hydraulic/pneumatic cylinder specifications required to safely elevate the platform under maximum payload conditions.
- **Base & Outrigger Stability:** Analyzed the center of gravity and tipping moments to ensure the lift remains completely stable during operation in rough or industrial terrain.

4. Economic Viability & Strategic Analysis

A crucial constraint of this project was the "Low-Cost" requirement. A structurally sound design is only successful if it can be manufactured and sold competitively.

Cost Analysis

Conducted a detailed breakdown of raw material costs (e.g., structural steel for the linkages, base framing), actuator components, and estimated machining/welding expenses. By optimizing the geometry of the scissor arms, I was able to reduce excess material weight, directly driving down the projected manufacturing cost.

SWOT Analysis

Executed a strategic **SWOT (Strengths, Weaknesses, Opportunities, Threats)** analysis to evaluate the proposed design's position in the industrial equipment market, anticipating potential manufacturing challenges and identifying target industry niches.

5. Academic Outcomes & Individual Achievement

Mastery of Mechanical Design

This individual project showcased my ability to independently execute a full engineering design cycle from initial market research to final mathematical validation. By successfully completing the **ME2851** project, I demonstrated cross-disciplinary competence—applying core mechanical engineering principles (statics, dynamics, machine design) alongside my primary expertise in Materials Science. The ability to prove a concept's viability purely through rigorous calculations, cost analysis, and structured reporting is a vital skill for high-level engineering roles.

Project Title: Quantum Mechanics of Tunneling Field Effect Transistors (TFETs)

Academic Module: MT2181 - Solid State Materials

Student Name: PREMAKUMARA H.P.S. (Index: 210494D) | **Team:** Group A

Project Nature: Theoretical Device Physics & Quantum Explanation

1. Executive Summary

This project was a purely theoretical research and presentation exercise designed to explain the complex device physics of **Tunneling Field Effect Transistors (TFETs)** using the fundamental principles of **Quantum Mechanics**. Unlike conventional engineering projects focused on fabrication, this assignment required a deep dive into solid-state physics to explain *why* the miniaturization of standard MOSFETs is reaching its physical limits, and *how* harnessing quantum tunneling can overcome the thermionic emission barrier (the 60 mV/decade sub-threshold swing limit).

2. Quantum Mechanical Foundations

The WKB Approximation

The core of the presentation utilized the Wentzel-Kramers-Brillouin (WKB) approximation to mathematically model the probability of an electron tunneling through the energy barrier. We explained how the transmission probability $T(E)$ is exponentially dependent on the effective mass of the carrier (m^*) and the bandgap energy (E_g) of the material.

Band-to-Band Tunneling (BTBT)

Unlike MOSFETs, where electrons must gain enough thermal energy to "jump over" the potential barrier (governed by the Fermi-Dirac distribution), TFETs operate by modulating the energy bands so electrons can tunnel directly from the valence band of the source to the conduction band of the channel.

Overcoming the Boltzmann Tyranny

We theoretically demonstrated that because TFETs filter out high-energy electrons (acting as an energy bandpass filter), they are not bound by the thermal limit of 60 mV/decade at

room temperature. This allows for an exceptionally steep sub-threshold swing, drastically reducing the off-state leakage current (I_{off}) and allowing the device to operate at highly scaled supply voltages.

3. Material Science in TFET Design

The project heavily evaluated how specific solid-state material properties dictate quantum tunneling efficiency.

Direct vs Indirect Bandgaps

High-k Dielectrics

Heterojunctions

Nanowire TFETs

Carbon Nanotubes (CNT)

- **Direct vs. Indirect Bandgaps:** Explained the necessity of momentum conservation (phonons) in indirect bandgap materials (like Silicon), which inherently lowers the tunneling probability compared to direct bandgap materials (like III-V compounds).
- **Heterojunction TFETs:** Analyzed architectures combining different semiconductor materials (e.g., Ge source and Si channel) to artificially lower the effective bandgap precisely at the tunneling junction, thereby exponentially increasing the ON current (I_{on}).
- **Carbon Nanotube (CNT) TFETs:** Explored futuristic 1D architectures where the ultra-short tunneling distance and unique density of states provide supreme control over ambipolar leakage and short-channel effects.

4. Advanced Electrical Characteristics

Ambipolar Behavior

A critical theoretical challenge we addressed is the ambipolar nature of TFETs, where parasitic tunneling occurs at the drain-channel junction under high drain biases. We explained material engineering solutions, such as asymmetric doping (lowly doped drain) or gate-underlap structures, to suppress this unwanted hole tunneling.

Drain Induced Barrier Lowering (DIBL)

We contrasted TFETs against MOSFETs in terms of short-channel effects, explaining how the tunneling mechanism inherently provides better immunity to DIBL, making it a superior candidate for sub-10nm transistor nodes.

5. Academic Outcomes

Mastery of Solid State Physics

This purely theoretical presentation proved my ability to bridge the gap between abstract quantum mechanics and practical semiconductor engineering. By successfully explaining

complex phenomena such as the WKB approximation, Fermi-Dirac statistics, and Band-to-Band Tunneling, I demonstrated a profound understanding of Solid State Materials. It highlights my capability to conduct high-level theoretical research and explain cutting-edge nanoscale device physics.

Project 1: Semester 5

Module Grade: A (2.0 C)

Project Title: Automotive Gearbox Design & Mechanical Optimization

Academic Module: ME3813 - Machine Design

Student Name: PREMAKUMARA H.P.S. (Index: 210494D) | **Team:** Group 11

Target Vehicle: Toyota Highlander 2024 (2050 kg, 5-Forward/1-Reverse Transmission)

1. Executive Summary

Conducted under the **ME3813 - Machine Design** module, this rigorous engineering project involved the complete mechanical and kinematic design of a manual transmission gearbox for a heavy SUV (Toyota Highlander 2024). The project required translating raw vehicle parameters—such as aerodynamic drag (0.34 Cd), mass (2050 kg), and gradient resistance (up to 30% inclines)—into precise mechanical components. The design encompassed gear ratio optimization, shaft fatigue analysis, bearing selection, and material metallurgy, proving cross-disciplinary competence in both mechanical design and materials engineering.

2. Kinematics & Gear Ratio Optimization

Traction & Power Calculations

- Calculated total tractive effort required to overcome rolling resistance, aerodynamic drag, and gradient forces on both flat roads and steep 30% inclines.
- Determined the engine torque and power curves required to achieve a top speed of 180 km/h while maintaining sufficient low-end torque for a 35 km/h climb on inclines.

Transmission Sequencing

- Engineered the exact gear ratios for a **5-speed forward and 1-speed reverse** transmission using geometric progression.
- Calculated the optimum RPM ranges to ensure smooth shifting without dropping below the engine's power band, balancing fuel economy (overdrive) with acceleration.

3. Gear Engineering & Metallurgy

Material Selection: SAE 5115 Case-Hardened Steel

Leveraging my background in Materials Science, **SAE 5115 steel** was selected for the

transmission gears. The case-hardening process provides a highly wear-resistant surface capable of withstanding extreme contact stresses, while maintaining a tough, ductile core to absorb shock loads and prevent brittle fracture during sudden torque spikes.

Rigorous Gear Calculations:

- **Load Analysis:** Calculated the tangential driving loads, dynamic loads (using Buckingham's equation), and static tooth strength (using the Lewis formula) for every gear pair.
- **Wear Resistance:** Conducted surface wear load calculations to ensure the gear flanks would not succumb to pitting or spalling over the vehicle's lifespan.
- **Geometry:** Determined exact module dimensions, center distances, and teeth counts to fit within a compact transmission housing.

4. Shaft Design & Fatigue Analysis

The transmission shafts (input, counter, and output) are subjected to complex, combined bending and torsional stresses. Designing them required strict adherence to the **ASME Code for Design of Transmission Shafting**.

Material & Stress Analysis

AISI 4340 High-Strength Alloy Steel was selected for the shafts due to its exceptional deep-hardenability and fatigue resistance. Maximum bending moments (from gear separation forces) and torsional moments were calculated to determine the minimum safe diameter for each shaft segment.

Fatigue Failure Prevention

Because shafts undergo fully reversed bending, dynamic fatigue analysis was critical. I applied failure criteria such as the **Soderberg and Goodman equations** to ensure the shafts had an infinite theoretical fatigue life under fluctuating torque loads.

5. Auxiliary Components & Tribology

SKF Deep Groove Ball Bearings

SKF Angular Contact Bearings

Spline Design

Keyway Stress Analysis

SAE 75W-90 Gear Oil

- **Bearing Selection:** Calculated the Equivalent Dynamic Bearing Load (W_{OR}) incorporating axial and radial thrust factors. Selected SKF Deep Groove and Angular Contact bearings, verifying their basic dynamic load rating (C) against a targeted lifespan (L_{10}) of millions of revolutions.
- **Power Transmission Connections:** Designed customized splines and keys, calculating the shear stresses and crushing (compressive) stresses to prevent hub failure at peak torque.
- **Lubrication Strategy:** Specified a dual Splash & Pressure lubrication system utilizing **SAE 75W-90 synthetic gear oil** to manage extreme pressure (EP) boundaries and dissipate thermal energy.

6. Academic Outcomes

Cross-Disciplinary Engineering Excellence

Earning an 'A' grade in ME3813 - Machine Design is a testament to my ability to master heavy Mechanical Engineering disciplines despite being rooted in the Materials Science

department. This project demanded the synthesis of complex physics (kinematics, statics, dynamics) with advanced metallurgy. Designing a complete, mathematically validated 5-speed automotive gearbox proves my capability to engineer robust, high-performance mechanical systems from the ground up.

Project Title: X-Ray Diffraction Analysis of Historical Paper: Deacidification & Ageing

Academic Module: MT3054 - Characterization of Materials

Student Name: PREMAKUMARA H.P.S. (Index: 210494D)

Focus Area: Cellulose Crystallinity & Nanostructural Preservation

1. Executive Summary

Conducted for the **MT3054 - Characterization of Materials** module, this individual research case study explored the nanostructural degradation of archival documents. While traditional historical preservation focuses on macroscopic properties (pH, color, tear strength), this project utilized advanced **X-Ray Diffraction (XRD)** to quantify how artificial ageing and aqueous deacidification treatments alter the fundamental crystallinity of cellulose fibers. The study compared 19th-century flax "rag" paper against modern cotton Whatman paper to determine how chemical conservation methods impact long-term material survivability.

2. Advanced Characterization Methodology (XRD)

Quantifying Cellulose Structure

To evaluate the structural shifts from amorphous to crystalline domains within the paper, XRD data was processed to calculate two critical metrics using 1D linear traces extracted from 2D diffraction images:

$$\text{Crystallinity Index (CI)} = (I_{\text{max}} - I_{\text{min}}) / I_{\text{max}}$$

$$\text{Crystallinity Ratio (CR)} = 1 - (I_{\text{min}} / (I_{\text{max}} - I_{\text{min}}))$$

By tracking the shifts in these indices, the study successfully quantified microscopic embrittlement that is entirely invisible to the naked eye.

3. Experimental Treatment Protocols

Artificial Ageing

To simulate decades of environmental degradation within a condensed timeframe, the paper samples were subjected to an artificial aging process in a controlled environmental chamber at **80°C and 65% Relative Humidity (RH)** for a continuous period of two weeks.

Chemical Deacidification

Acid hydrolysis is the primary enemy of cellulose. The samples were treated with two standard archival neutralization agents to assess their nanostructural impact:

- Calcium Hydroxide $\text{Ca}(\text{OH})_2$
- Magnesium Bicarbonate $\text{Mg}(\text{HCO}_3)_2$

4. Key Findings & Diffraction Analysis

The XRD analysis revealed highly specific interactions between the conservation chemicals and the cellulose matrix:

- **Ageing-Induced Crystallinity:** The artificial aging process demonstrably increased the crystallinity of the cellulose fibers. As the amorphous regions degraded faster under thermal stress, the relative crystalline fraction increased, leading to macroscopic embrittlement of the paper.
- **Calcite Precipitation:** The XRD diffraction patterns for samples treated with Ca(OH)_2 revealed distinct residual calcite reflections. While this acts as an alkaline reserve, the crystalline calcite deposits disrupt the native nanostructure, potentially compromising the paper's mechanical integrity over time.
- **Treatment Viability:** The study concluded that while deacidification prevents acid-catalyzed scission, improper treatments can inadvertently cause excessive crystallization, proving that chemical conservation is a delicate balance.

5. Academic Outcomes

Mastery of Material Characterization

This case study perfectly encapsulated the objectives of the **MT3054 - Characterization of Materials** module. It demonstrated my ability to apply complex analytical techniques—traditionally associated with hard metallurgy or ceramics—to soft, organic polymeric structures like cellulose. By successfully interpreting 2D XRD images, isolating specific elemental reflections (like Calcite), and calculating quantitative structural indices (CI and CR), I proved my capability to perform high-level forensic material analysis and non-destructive testing.

Engagement: 24-Week Industrial Training / Internship

Company: Consulting Engineering & Contractors (Pvt) Ltd. (CEC)

Project: RDA Road Rehabilitation & Improvement (Homagama Div. Secretariat)

Role: Trainee Material Engineer

1. Executive Summary

During Semester 6, I completed a rigorous 24-week industrial training program as a **Trainee Material Engineer** for Consulting Engineering & Contractors (Pvt) Ltd. The core focus of the internship was the rehabilitation, improvement, and maintenance of rural road infrastructure under the supervision of the Road Development Authority (RDA) and Euro Group Consultants. This placement provided extensive hands-on experience in civil materials testing, asphalt plant management, quality assurance, and site safety protocols, bridging the gap between theoretical materials science and large-scale industrial application.

2. Asphalt Plant Operations & Production

Mastery of Asphalt Mixing Technology

A major component of my training involved understanding and managing the complex machinery used in road paving. I gained deep technical exposure to the operational mechanics and environmental impacts of various Asphalt Plants.

Plant Classifications

- **Batch vs. Drum:** Analyzed the precision of Batch Plants (mixing weighed aggregates and bitumen in cycles) versus the continuous, high-volume output of Drum Plants.
- **Thermal Flow:** Evaluated Parallel Flow (aggregates and gases flow in the same direction) vs. Counterflow systems (opposite directions for superior heat transfer and reduced emissions).

Sustainable Paving

- **RAP Integration:** Studied the utilization of Reclaimed Asphalt Pavement (RAP) in Double Barrel Drum Plants to reduce raw material costs and lower the carbon footprint of construction.
- **Eco-Friendly Plants:** Assessed the implementation of dust filters and enclosed emission controls to meet strict environmental regulations.

3. Quality Assurance & Materials Testing

As a Trainee Material Engineer, ensuring the structural integrity of the road layers was my primary responsibility. All testing and quality control procedures were conducted in strict adherence to international engineering standards.

ASTM Standards

BS Standards

Aggregate Testing

Bitumen Analysis

Soil Compaction

- **Laboratory and Field Testing:** Executed physical and chemical tests on raw materials (soil, aggregates, and bitumen) to ensure they met the rigorous specifications set by the RDA.
- **Data Synthesis:** Synthesized test data to provide valid conclusions on material suitability, preventing structural failures such as rutting or cracking in the final pavement.

4. Professional Development & Program Outcomes (POs)

Engineering & Society

Directly applying the concepts learned in my **MT3880 - Engineer & Society (A+)** module, I assessed the societal and environmental impacts of road construction. This included implementing Health, Safety, and Environmental (HSE) practices to protect both site workers and Inspectors) to understand resource and finance the local Homagama community.

Ethics & Project Management

Practiced strict professional ethics by ensuring completely accurate, untempered test reporting. Additionally, I observed the complex hierarchical project management structure (interacting with Project Managers, Quantity Surveyors, and QA allocation).

5. Academic Outcomes

Transitioning to a Professional Engineer

This 24-week internship successfully fulfilled the core requirements of **MT3993 - Industrial Training (A-)**. It allowed me to achieve all 12 Washington Accord Program Outcomes (POs)—from complex problem analysis and modern tool usage to communication, teamwork, and lifelong learning. By taking ownership of material quality on a major government infrastructure project, I transformed my academic knowledge of polymers, ceramics, and ferrous metals into practical, large-scale civil engineering competence.

Project Title: Designing a Corrosion-Resistant Marine Propulsion Shaft
Academic Module: MT3201 - Comprehensive Design Project
Student Name: PREMAKUMARA H.P.S. (Index: 210494D) | **Team:** Group 01
Primary Role: Lead Material Selector (Ansys Granta) & FEA Analyst (Abaqus)

1. Executive Summary

Serving as the pinnacle of the engineering undergraduate degree, the **MT3201 Comprehensive Design Project** required the complete, from-scratch engineering of a 1.6m marine propulsion shaft coupled to a CAT 3406C marine engine (298 kW / 1800 RPM). Operating in a highly aggressive seawater environment, the shaft demanded supreme resistance to crevice corrosion, pitting, and catastrophic torsional fatigue over a targeted 20-year lifespan. This project synthesized every aspect of materials science—from macroscopic mechanical geometry to microscopic metallurgical selection.

2. Advanced Material Selection via Ansys Granta

Computational Materials Engineering

My core responsibility was to navigate thousands of engineering alloys to find the perfect balance of yield strength, cost, and extreme marine durability. To do this, I utilized **Ansys Granta Selector**, generating highly specific Ashby selection charts.

Selection Parameters & Constraints

- **Mechanical Constraint:** Yield strength strictly > 400 MPa to handle the massive 1.58 kN·m continuous torque.
- **Chemical Constraint:** Pitting Resistance Equivalent Number (PREN) > 40 to ensure immunity against chloride-induced localized corrosion.

The Elimination Process

I plotted candidate families including Monel K-500, Titanium Alloys, and various Stainless Steels. While Titanium offered perfect corrosion resistance, the cost penalty was too high. Austenitic steels (like 316L) failed the yield strength criteria for the required shaft diameter.

The Final Material: Super Duplex Stainless Steel (UNS S32750 / SAF 2507). Its two-phase microstructure (50% Austenite / 50% Ferrite) provided the exact solution: the high strength of ferrite combined with the excellent ductility and localized corrosion resistance of austenite.

3. Finite Element Analysis (FEA) using Abaqus

Material properties on paper mean nothing without structural validation. My secondary core responsibility was to digitally validate the mechanical geometry of the Super Duplex shaft under dynamic loading using **Abaqus FEA simulation software**.

Abaqus CAE

Torsional Shear Stress

Von Mises Yield Criterion

Stress Concentrations

Mesh Refinement

- **Load Application:** Modeled the 1.6m solid shaft and applied the calculated maximum engine torque of 1.58 kN·m alongside the axial thrust generated by the propeller.
- **Stress Evaluation:** Ran Von Mises stress simulations to ensure the combined torsional and bending stresses remained well below the 550 MPa yield strength of the Super Duplex steel, incorporating a rigorous safety factor for dynamic marine loading.
- **Fatigue Optimization:** Identified critical stress concentrations (at keyways and flanges) and iteratively smoothed the CAD geometry radii in Abaqus to prevent fatigue crack initiation points.

4. Holistic Systems Engineering: Active Corrosion Protection

Despite the inherent resistance of Super Duplex steel, standard marine classification societies (DNV/ABS) require active fail-safes. Working with the team, I helped evaluate the integration of an **Impressed Current Cathodic Protection (ICCP)** system utilizing an MMO Anode.

Furthermore, because the oil film inside the bearings electrically isolates the shaft from the hull's cathodic protection, we designed a highly specific **Shaft Earthing Device (Silver-Graphite Slip Ring)**. This provided a low-resistance path ($< 0.01 \Omega$) to ground, effectively preventing devastating "Spark Erosion" on the main engine bearings.

5. Academic Outcomes

Pinnacle Engineering Achievement

Scoring an **A+ (4.0 GPA)** in the Comprehensive Design Project is a definitive proof of competence. By spearheading the highly technical **Ansys Granta material selection** and the **Abaqus FEA simulations**, I demonstrated the ability to lead the most critical phase of a major industrial project: ensuring the product will not physically fail under extreme, real-

world stresses. This project bridged the gap between theoretical metallurgy and advanced computational engineering.

Project 2: Semester 7

Individual FEA Project

Project Title: Finite Element Analysis of Spur Gear Contact Mechanics

Academic Module: MT4810/MT4811 - FEA Design Project (Continuum Scale)

Student Name: PREMAKUMARA H.P.S. (Index: 210494D)

Primary Tools: Abaqus CAE, Solid Edge, Hertzian Contact Theory

1. Executive Summary

Conducted as an individual continuous assessment for the **MT4810/MT4811 Continuum Scale** modules, this project focused on simulating the complex tribological and mechanical interactions between a spur gear and pinion. The objective was to utilize advanced Finite Element Analysis (FEA) to evaluate the stress distribution, plastic deformation, and contact mechanics under high torsional loads. The project required parametric CAD modeling, rigorous boundary condition setup, and a mathematical mesh convergence study to prove the reliability of the computational results.

2. Parametric CAD Modeling & Pre-Processing

Index-Driven Dimensional Constraints

The geometry of the transmission system was uniquely personalized based on my specific student index number (210494D). I derived the following critical dimensions using the last three digits (\$X = 494\$):

Gear Pitch Circle Diameter (PCD) = 494 mm

Pinion Pitch Circle Diameter (PCD) = $494 / 5 = 98.8$ mm

Applied Torsional Load = 494 N.m (Torque)

Geometric Generation

Using **Solid Edge**, I accurately modeled the involute tooth profiles for both the gear and the pinion to ensure theoretically perfect meshing. The components were assembled at the exact center distance $(D_G + D_P)/2$ to prevent interference or excessive backlash.

STEP File Topology Transfer

The assembly was exported as a precise STEP file into Abaqus. This ensured that the complex involute curves (surfaces, curves, volumes) transferred accurately as frozen bodies, preparing them for advanced meshing without topology distortion.

3. FEA Setup & Continuum Mechanics (Abaqus)

Simulating dynamic metal-to-metal contact requires highly specific boundary conditions and material definitions to prevent simulation failure (non-convergence).

Surface-to-Surface Contact

Coulomb Friction ($\mu = 0.1$)

Elasto-Plasticity

Encastre Constraints

C3D10 / C3D8R Elements

- **Material Definition:** Assigned **Stainless Steel** properties. Crucially, I inputted both the linear elastic modulus and the **non-linear plasticity data** (yield strength, hardening curve) to accurately capture permanent deformation at the tooth roots.
- **Contact Interactions:** Established a Surface-to-Surface contact formulation with a penalty friction coefficient of 0.1. The Pinion was driven, and the Gear was subjected to the 494 units of torque.
- **Post-Processing Extraction:** Extracted critical contour plots for **Von Mises Stress** (to evaluate yielding), **Equivalent Plastic Strain (PEEQ)**, and **Overall Displacement** at the contact interface.

4. Mesh Convergence & Theoretical Validation

Mesh Convergence Study

To prove the mathematical accuracy of the simulation, I conducted a rigorous convergence study. I iteratively reduced the global mesh size (\$5mm \rightarrow 3mm \rightarrow 2mm \rightarrow 1mm\$) using tetrahedral (C3D10) and hexahedral (C3D8R) elements. By plotting Maximum Von Mises Stress vs. Number of Elements, I demonstrated the exact point where the stress values stabilized, proving the results were independent of the mesh grid.

Hertzian Contact Validation

As an advanced validation step, I compared the peak contact stresses generated by the Abaqus simulation against classical **Hertzian Contact Stress calculations**. Proving that the computational output matched the theoretical hand-calculations elevated the project to a top-tier engineering standard.

5. Academic Outcomes

Mastery of Computational Solid Mechanics

This individual project perfectly showcased my ability to handle continuum scale simulations. It proved that I can move beyond basic CAD and perform high-level structural integrity assessments. By successfully navigating Abaqus interaction constraints, material non-linearity, and mesh convergence, I demonstrated the exact skills required by modern R&D industries for mechanical design validation and failure analysis.

Project 3: Engineering Economics

Group Project

Project Title: Viability Analysis of Ferrocement Wall Panels for Construction

Academic Module: MN4023 - Engineering Economics

Student Name: PREMAKUMARA H.P.S. (Index: 210494D) | **Team:** Group Project

Core Focus: Financial & Economic Benefit-Cost Analysis (FBCA & EBCA)

1. Executive Summary

Conducted for the **MN4023 - Engineering Economics** module, this project evaluated the commercial and societal viability of manufacturing prefabricated **Ferrocement Wall Panels**. Aimed at replacing traditional brick and block walls, these panels utilize a cement-sand mortar reinforced with wire mesh to drastically reduce construction time and material costs. Working in collaboration with National Engineering Research & Development Centre (NERDC) prototypes, our team performed rigorous financial modeling, economic shadow-pricing, and sensitivity analyses to prove the project's feasibility for large-scale implementation.

2. Technical Feasibility & Design Optimization

Before assessing the financial metrics, the structural integrity of the panels had to be validated to ensure market acceptance:

- **Structural Simulation:** Conducted linear-elastic static analysis using **COMSOL Multiphysics** to evaluate the self-weight deflection of the panels when simply supported at short edges.
- **Material Composition:** Optimized the mixture utilizing River Sand, M-sand, Fly Ash, and mild steel wire mesh to reduce the excessive self-weight and mitigate the cracking issues present in early NERDC prototypes during lifting and handling.

3. Financial Benefit-Cost Analysis (FBCA)

Private Sector Commercial Viability

The FBCA evaluated the project from the perspective of a private investor, utilizing market

prices to determine profitability. Assuming a Minimum Acceptable Rate of Return (MARR) of 15% and a 5-year project lifespan, the investment yielded highly favorable metrics.

Net Present Value (NPV)

12.85 Million LKR

(NPV > 0 = Accept)

Internal Rate of Return (IRR)

22%

(IRR > 15% MARR)

Benefit-Cost Ratio (BCR)

1.09

(BCR > 1 = Feasible)

Discounted Payback Period: 4.2 Years

4. Economic Benefit-Cost Analysis (EBCA)

While the FBCA proved commercial profitability, the EBCA evaluated the project from a **macro-economic, national perspective**. This required converting market prices to shadow prices by stripping away market distortions such as taxes and subsidies, and quantifying external social impacts.

Quantified Social Benefits

- **Environmental Impact:** Significant reduction in \$CO_2\$ emissions compared to traditional fired-clay brick manufacturing.
- **Socio-Economic:** Creation of direct local employment opportunities, reskilling of traditional masonry workers, and the facilitation of affordable, rapid housing infrastructure.

Economic Metrics

Because the social benefits heavily outweighed the drawbacks (such as minor labor displacement), the economic indicators were even stronger than the financial ones:

- **Economic NPV:** 19.41 Million LKR
- **Economic IRR:** 27%
- **Economic BCR:** 1.15

5. Sensitivity & Risk Analysis

Stress-Testing the Investment

To ensure the robustness of our recommendation, we conducted a sensitivity analysis by simulating a **10% decrease in projected revenue**.

Even under this pessimistic scenario, the NPV remained positive (dropping from 12.8M to 1.7M LKR) and the IRR remained at 16% (still safely above the 15% MARR constraint). This proved that while the project is highly sensitive to market demand, it possesses a sufficient safety margin.

Risk Mitigation Strategies:

- **Technical Risks:** Mitigated panel cracking via rigorous COMSOL design optimization and protective coatings.
- **Market Risks:** Combated public reluctance toward new building materials via targeted awareness campaigns and the construction of pilot display houses.

6. Academic Outcomes

Mastering Commercial Engineering

This project represented a critical leap from purely technical engineering to high-level project management and economics. By successfully executing both FBCA and EBCA methodologies—calculating NPV, IRR, and BCR while accounting for shadow pricing and sensitivity—I demonstrated the ability to evaluate whether an engineering innovation is not just physically possible, but economically viable and socially beneficial for large-scale implementation.

Project Title: Synergistic Effects of Rice Husk Ash and Polypropylene Fiber on the Performance of Modified Expanded Polystyrene Cement Mortar

Academic Milestone: Final Year Research Project (FYP)

Student Name: PREMAKUMARA H.P.S. (Index: 210494D) | **Partner:** K. Mayoorathan

Execution Period: Semesters 6, 7, and 8 (Culminating in July 2026)

1. Executive Summary

Serving as the capstone of the B.Sc. Engineering degree, this comprehensive Final Year Project (FYP) focused on developing a novel, sustainable, lightweight construction material. While Expanded Polystyrene (EPS) cement mortar offers excellent thermal insulation and reduces the dead load of high-rise structures, it inherently suffers from low mechanical strength and high brittleness. This research successfully engineered a highly optimized composite matrix by introducing **Rice Husk Ash (RHA)** as a pozzolanic binder and **Polypropylene (PP) Fibers** as structural reinforcement, neutralizing the weaknesses of traditional EPS mortar while maximizing its sustainability.

2. The Material Triad: Core Innovations

Expanded Polystyrene (EPS)

Utilized as ultra-lightweight artificial aggregates. EPS dramatically reduces the density and thermal conductivity of the mortar but creates weak interfacial zones that act as stress concentrators and crack initiators.

Rice Husk Ash (RHA)

An agricultural waste product rich in amorphous silica. By replacing a percentage of Ordinary Portland Cement (OPC) with RHA, the pozzolanic reaction produces additional Calcium Silicate Hydrate (C-S-H) gel, significantly densifying the Interfacial Transition Zone (ITZ) around the EPS beads.

The Bridging Mechanism: Polypropylene (PP) Fibers

To counteract the inherent brittleness caused by the EPS beads, synthetic PP fibers were dispersed throughout the mortar. These fibers act as a microscopic secondary reinforcement system. When micro-cracks initiate under stress, the high tensile strength of

the PP fibers bridges the crack faces, transferring loads across the fracture and vastly improving the flexural toughness and ductility of the composite.

3. Experimental Methodology & Testing

The research demanded rigorous laboratory execution, from precise mix design formulation to advanced mechanical characterization following ASTM and Sri Lankan Standards (SLS).

Mix Design Optimization

Compressive Strength Testing

Flexural Strength Testing

Density & Workability Analysis

Pozzolanic Reactivity

- **Sample Preparation:** Cast multiple series of mortar cubes and prisms with varying volume fractions of EPS, RHA, and PP fibers to isolate the independent and synergistic variables.
- **Mechanical Evaluation:** Conducted 7-day and 28-day compressive and flexural strength tests. The data definitively mapped the trade-off between density reduction (via EPS) and strength retention (via RHA/PP synergy).
- **Failure Analysis:** Evaluated the post-peak failure modes, proving that the inclusion of PP fibers shifted the failure mechanism from sudden, catastrophic brittle fracture to a progressive, ductile yielding.

4. Sustainable Engineering Impact

Eco-Friendly Infrastructure

The cement industry is a massive contributor to global CO_2 emissions. By substituting energy-intensive Portland cement with RHA (an agricultural by-product), the project directly aligns with green engineering principles. Furthermore, utilizing EPS addresses plastic waste management.

Commercial Viability

The resulting composite is an ideal candidate for non-load-bearing applications in commercial real estate, such as pre-cast partition walls, acoustic barriers, and thermal insulation facades, significantly lowering foundation costs due to reduced dead weight.

5. Academic Outcomes

Culmination of the Materials Science Degree

The Final Year Project represents the transition from an engineering student to an independent researcher. Managing a project of this magnitude spanning three semesters required advanced synthesis of previous modules—including Polymer Engineering (EPS/PP), Ceramic/Cementitious Materials (RHA/C-S-H), and Mechanical Behavior of Materials

(fracture mechanics). This thesis stands as undeniable proof of my ability to innovate, execute rigorous scientific methodologies, and contribute valuable, sustainable solutions to the field of Materials Engineering.

Focus Areas: Energy Materials & Biomaterials

Academic Modules: MT4420 - Energy Materials & Biomaterials Elective

Student Name: PREMAKUMARA H.P.S. (Index: 210494D)

Scope: Device Simulation, Electrochemical Fabrication & Biological Interfaces

1. Executive Summary

During the final semester of the B.Sc. Engineering degree, my coursework diversified into highly specialized, cutting-edge material applications. This portfolio documents three distinct activities across two advanced domains: **Energy Materials** and **Biomaterials**. The projects ranged from the theoretical quantum simulation of Perovskite solar cells (SCAPS-1D) to the hands-on fabrication and electrochemical testing of Graphene Supercapacitors, alongside a comprehensive research presentation on the osseointegration mechanics of dental implants. This multidisciplinary exposure demonstrates a profound capability to engineer materials at the nexus of energy storage, renewable power, and human biology.

2. Biomaterials: Osseointegration of Dental Implants

The Biological-Material Interface

This research presentation focused on the critical mechanics of **Osseointegration**—the direct structural and functional connection between living bone tissue and load-bearing artificial implants (typically Titanium or Zirconia) without intervening fibrous layers.

Topographical Modifications

- **Macro to Nano Scale:** Evaluated techniques like Sandblasting and Large-grit Acid etching (SLA) to increase the implant's surface area.
- **Mechanics:** Demonstrated how these micro-rough surfaces promote initial mechanical interlocking (primary stability) and facilitate blood clot retention for rapid healing.

Bioactive Coatings

- **Chemical Integration:** Analyzed the application of **Hydroxyapatite (HA)** coatings via plasma spraying to mimic natural bone mineral.
- **Protein Incorporation:** Explored futuristic augmentations using BMP-2 proteins to actively stimulate osteoblast differentiation and accelerate bone deposition.

3. Energy Materials: SCAPS-1D Solar Cell Simulation

As part of the MT4420 Energy Materials module, I conducted an individual computational analysis on a next-generation **Perovskite Solar Cell (FTO/ZnO/X/Cu2O/Au)**.

SCAPS-1D

Power Conversion Efficiency (PCE)

Carrier Recombination

Fill Factor (FF)

Parametric Optimization

Using SCAPS-1D software, I simulated the effect of the perovskite absorber layer's thickness on critical photovoltaic parameters. I generated and analyzed continuous plots for Open-Circuit Voltage (V_{oc}), Short-Circuit Current Density (J_{sc}), Fill Factor (FF), and overall Power Conversion Efficiency (PCE).

Quantum Efficiency Analysis

I mathematically proved that while increasing thickness captures more photons (raising J_{sc} up to 29.3 mA/cm^2), exceeding $1.0 \mu\text{m}$ degrades the internal electric field. This forces charge carriers to rely on slow diffusion rather than drift, increasing recombination rates and destroying the Fill Factor. The optimum thickness was precisely pinpointed at $\sim 0.9 - 1.1 \mu\text{m}$.

4. Energy Materials: Graphene Supercapacitor Fabrication

Hands-On Electrochemical Engineering

Transitioning from software simulation to physical fabrication, this laboratory project involved synthesizing a **Graphene-based Supercapacitor Electrode**. The objective was to harness graphene's extraordinary theoretical surface area ($2630 \text{ m}^2/\text{g}$) to create an Electrical Double-Layer Capacitor (EDLC) capable of bridging the gap between high-energy batteries and high-power traditional capacitors.

- **Fabrication:** Synthesized the electrode matrix using active graphene powder bound with PTFE, uniformly coated onto a conductive substrate.
- **Cyclic Voltammetry (CV):** Conducted CV sweeps across multiple scan rates to evaluate the capacitive behavior. Interpreted the quasi-rectangular CV curves to confirm ideal EDLC charge-storage mechanisms without parasitic faradaic redox reactions.
- **Galvanostatic Charge-Discharge (GCD):** Executed GCD testing at various current densities. By analyzing the discharge curve slopes ($\Delta V / \Delta t$), I calculated the precise Specific Capacitance (C_s) of the graphene electrode, validating its high-power density applications.

5. Academic Outcomes

Versatility at the Bleeding Edge

This portfolio of Semester 8 activities proves my ability to operate across entirely different spectrums of modern materials science. Whether I am analyzing how human osteoblast cells adhere to titanium, simulating electron-hole pair recombination in perovskite lattices using SCAPS-1D, or physically fabricating nanocarbon energy storage devices, I possess the theoretical depth and practical laboratory skills to drive innovation in high-tech industries.

Engineering & Technical Portfolio

AI-Accelerated Development

Focus Area: Systems Architecture, IoT, & AI-Assisted Software Engineering

Developer: PREMAKUMARA H.P.S. (Index: 210494D)

Scope: Full-Stack Web Apps, Embedded Systems (C++), and Geotechnical Python Algorithms

1. Executive Summary

As a Materials Science & Engineering graduate, my approach to software and system development is grounded in **Systems Engineering** rather than raw manual coding. My extracurricular projects demonstrate an ability to identify a problem, design a structural/logic solution, and leverage modern AI tools to accelerate the implementation and deployment of those solutions. This portfolio showcases my capability as an **Engineering Architect** who seamlessly bridges the gap between hardware, materials, and code.

2. Full-Stack Systems & Web Architecture

AL Physics Academy LMS

Role: Systems Architect & Product Lead

Architecting a complete, serverless Learning Management System (LMS) for A/L Physics tuition to manage students, automated grading, and study materials.

Node.js / Express

Supabase (PostgreSQL)

Vercel Serverless

JWT Authentication

- Designed the relational database schema (PostgreSQL) and secure authentication flows (JWT/bcrypt).
- Leveraged AI to rapidly scaffold the Node.js/Express backend, ensuring secure role-based access and seamless file upload pipelines.

Family Money Manager App

Role: Developer & End-User

Developed a zero-cost, cloud-synced personal finance web application optimized for mobile browsers, currently used daily for dual-user income and expense tracking.

HTML/CSS/JS

Google Apps Script

Cloud Architecture

- Architected a highly responsive front-end UI/UX using HTML/CSS/JS with a focus on cross-platform accessibility.
- Connected the frontend to a Google Sheets backend via custom API scripting, utilizing AI to optimize asynchronous data fetching.

3. Python Algorithms for Geotechnical Analysis

Dynamic Cone Penetrometer (DCP) Layer Analyzer

Developed during my industrial internship, this tool automates the complex geotechnical analysis of soil layer boundaries from raw field data, replacing manual graph plotting with instant, high-precision mapping.

- **Mathematical Optimization:** Engineered a Python-based algorithm using **Root Mean Square Deviation (RMSD)** combined with linear interpolation to detect significant shifts in soil resistance.
- **AI-Assisted Implementation:** Architected the core mathematical logic and utilized AI to refine the data processing loops and array handling, rapidly deploying a functional tool that significantly improved site investigation speed.

4. Embedded Systems & Internet of Things (IoT)

Adaptive AC Blinking System

Engineered an advanced, Wi-Fi-controlled smart relay system designed to manipulate standard AC bulbs with adaptive blinking patterns not available in commercial markets.

ESP8266

C++ Firmware

OTA Updates

- Designed the hardware relay controller and safety isolation for AC loads.
- Leveraged AI to structure the embedded HTTP web server and Over-The-Air (OTA) firmware update logic for remote maintenance.

Autonomous Arduino Car

A hobbyist robotics project integrating sensor arrays with motor drive logic to create a self-navigating vehicle.

Arduino UNO

HC-SR04 Ultrasonics

Motor Drive Logic

- Programmed complex reactive logic for obstacle avoidance, including a timestamp-based "Stuck Detection" mechanism.
- Utilized AI to accelerate the C++ firmware writing, focusing my efforts on system logic, sensor integration, and calibration.

5. The Cross-Disciplinary Engineer

The 10x Engineering Multiplier

These extracurricular projects prove my ability to operate as a modern Systems Architect. By combining deep domain knowledge in materials and mechanics with the ability to define logical software solutions, I use AI as a force multiplier to build and deploy complex tools—from geotechnical Python scripts to cloud-connected IoT devices—at exceptional speed.

SANDUN PREMAKUMARA

B.Sc. Engineering Graduate | Independent Physics Educator & Consultant

1. The Educator & Technical Communicator

Over the past five years, I have independently managed and conducted advanced group consulting classes for G.C.E. Advanced Level (A/L) Physics students. Balancing this responsibility with a rigorous 161-credit Materials Science and Engineering curriculum at the University of Moratuwa has required immense discipline, time management, and unwavering dedication to both my academic pursuits and my students' success.

This dual role has honed my leadership and advanced technical communication skills. I specialize in consistently breaking down complex physical, mathematical, and mechanical concepts into easily understandable, actionable lessons. Whether explaining thermodynamic cycles or quantum mechanics, my objective is to translate complex theories into practical knowledge that students can grasp and apply.

"I do not just teach Physics for exams; my goal is to build an Engineering Mindset in every student."

2. "Physics with Tech" - The Premium Teaching Concept

My approach to teaching heavily integrates modern technology, setting my consulting practice apart from traditional tutoring methods. I leverage my background in software and systems architecture to create a "Premium Value" experience for my students:

- **Interactive Simulations:** Instead of relying solely on a whiteboard to explain challenging theoretical segments (e.g., projectile motion, electromagnetic induction), I utilize software simulations directly in the classroom. This visual, data-driven approach allows students to see physics in action, transforming abstract equations into tangible mechanics.
- **The A/L Physics Academy LMS:** Taking initiative beyond standard classroom hours, I developed and managed an A/L Physics Academy Learning Management System (LMS). This digital platform enables structured, asynchronous learning and resource distribution, showcasing my ability to build and deploy educational tech tools.
- **AI as an Educational Force Multiplier:** I actively incorporate AI tools to generate custom problem sets, dynamic demonstrations, and personalized learning pathways, ensuring my teaching methods remain at the cutting edge of educational technology.

3. Bridging Engineering, Leadership, and Values

Technical Mentorship

My experience leading these classes proves my capability as a self-starter. Instructing students for five years demonstrates my capacity to mentor others, manage group dynamics, and adapt my communication style to different learning paces—essential traits for any engineering leadership role.

Mindfulness in Tech

Beyond technical rigor, I integrate personal values into my teaching. By balancing the study of Dhamma with modern technological advancement, I promote "Mindfulness in Tech." This holistic philosophy encourages students to pursue engineering with a balanced, ethical, and focused mindset.

4. Core Competencies Gained

- **Complex Problem Translation:** Ability to simplify heavy mathematical and mechanical concepts.
- **Public Speaking & Presentation:** Five years of continuous verbal communication, presenting to and engaging groups of students.
- **Ed-Tech Development:** Experience in deploying digital tools (LMS) to solve logistical and educational bottlenecks.
- **Project & Time Management:** Successfully maintaining high academic standards (Second Class Lower) while running an independent educational venture.